# The Short Run Trade-off between Inflation and Unemployment

# Introduction

- Inflation on long run determined by money supply
- Unemployment in long run determined by population growth, and other factors like efficiency wages, unions, job search frictions etc.
- Thus, no long run relationship between the two.
- In short run however, society faces a short run tradeoff between inflation and unemployment.

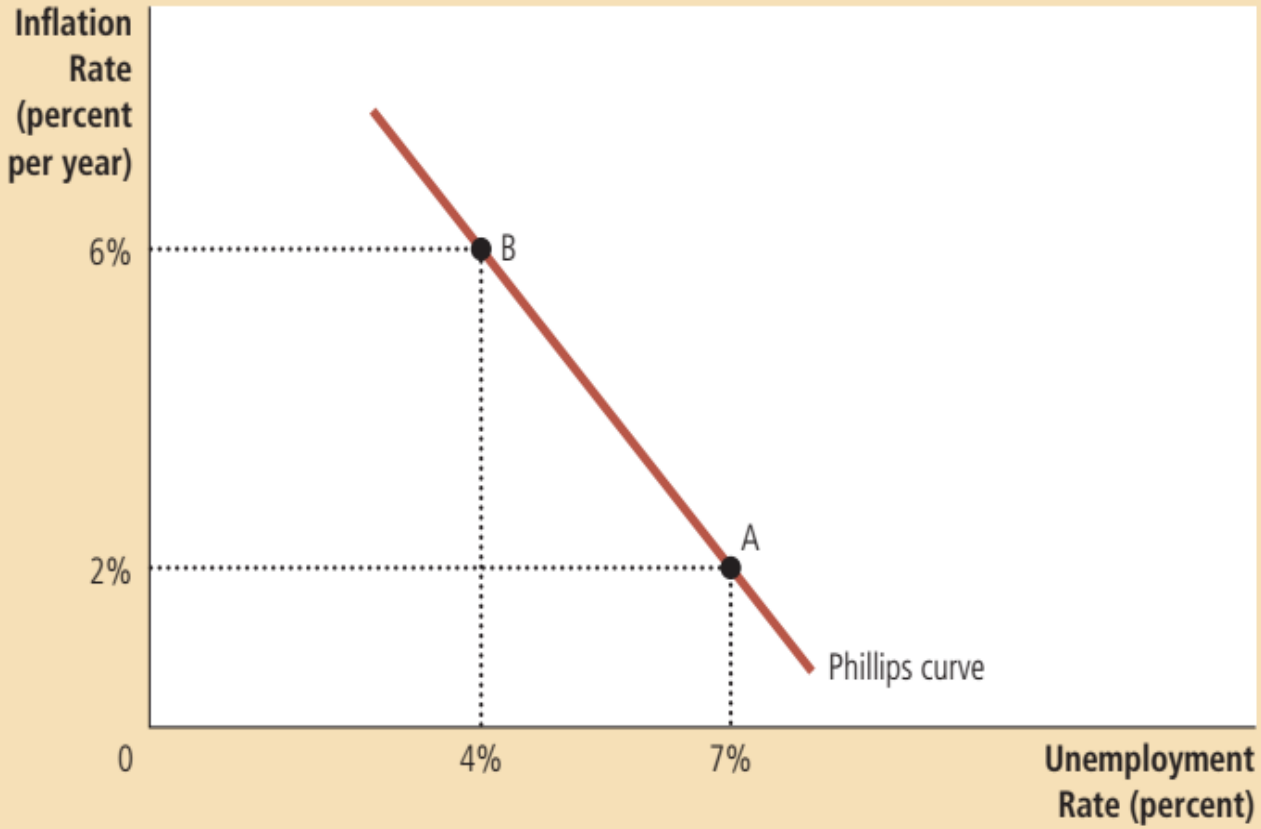



Figure 1

**The Phillips Curve**

The Phillips curve illustrates a negative association between the inflation rate and the unemployment rate. At point A, inflation is low and unemployment is high. At point B, inflation is high and unemployment is low.

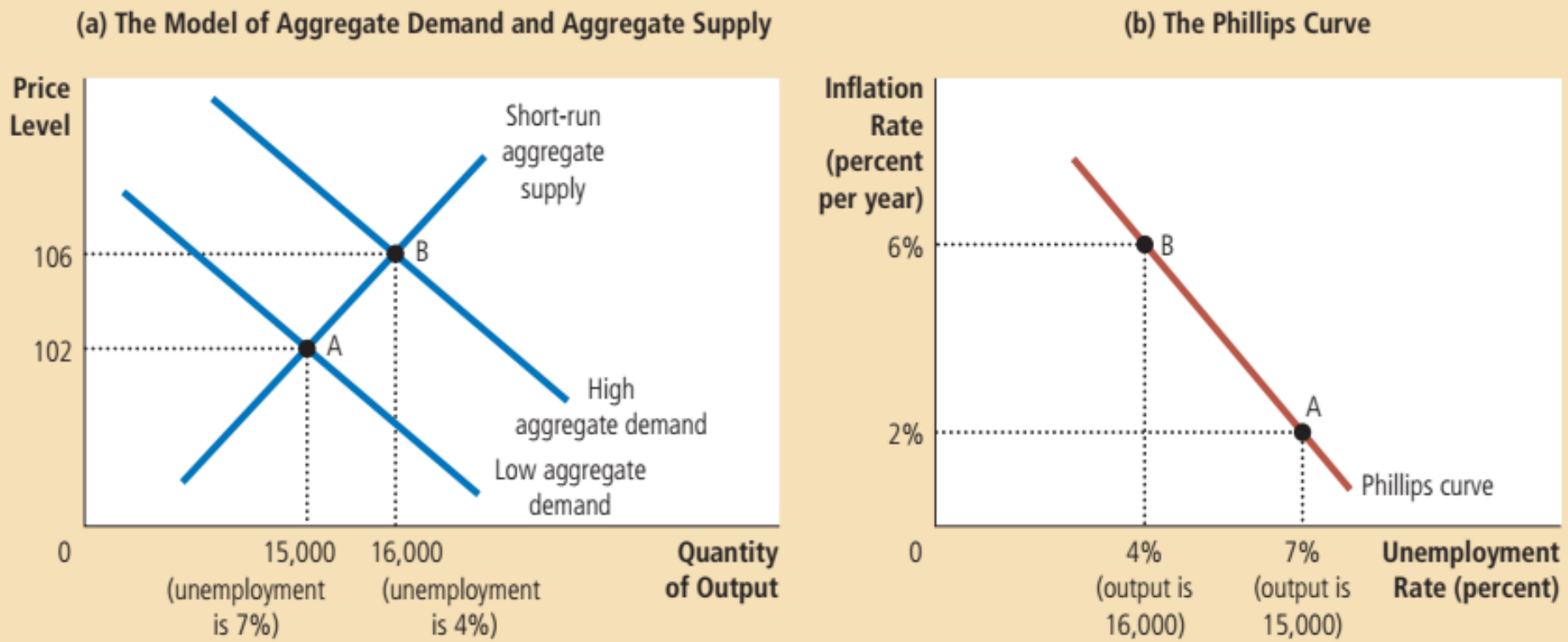
# AD-AS and PC

- The menu of choices from Phillips Curve (PC) can be derived from the AD-AS model.
- Higher AD leads to higher demand for goods and services, leading to higher output, lower unemployment, but also higher prices (growth).
- Expansionary MP and FP lead to higher AD, hence they can move an economy **along** the PC.

Figure 2

**How the Phillips Curve Is Related to the Model of Aggregate Demand and Aggregate Supply**

This figure assumes a price level of 100 for the year 2020 and charts possible outcomes for the year 2021. Panel (a) shows the model of aggregate demand and aggregate supply. If aggregate demand is low, the economy is at point A; output is low (15,000), and the price level is low (102). If aggregate demand is high, the economy is at point B; output is high (16,000), and the price level is high (106). Panel (b) shows the implications for the Phillips curve. Point A, which arises when aggregate demand is low, has high unemployment (7 percent) and low inflation (2 percent). Point B, which arises when aggregate demand is high, has low unemployment (4 percent) and high inflation (6 percent).



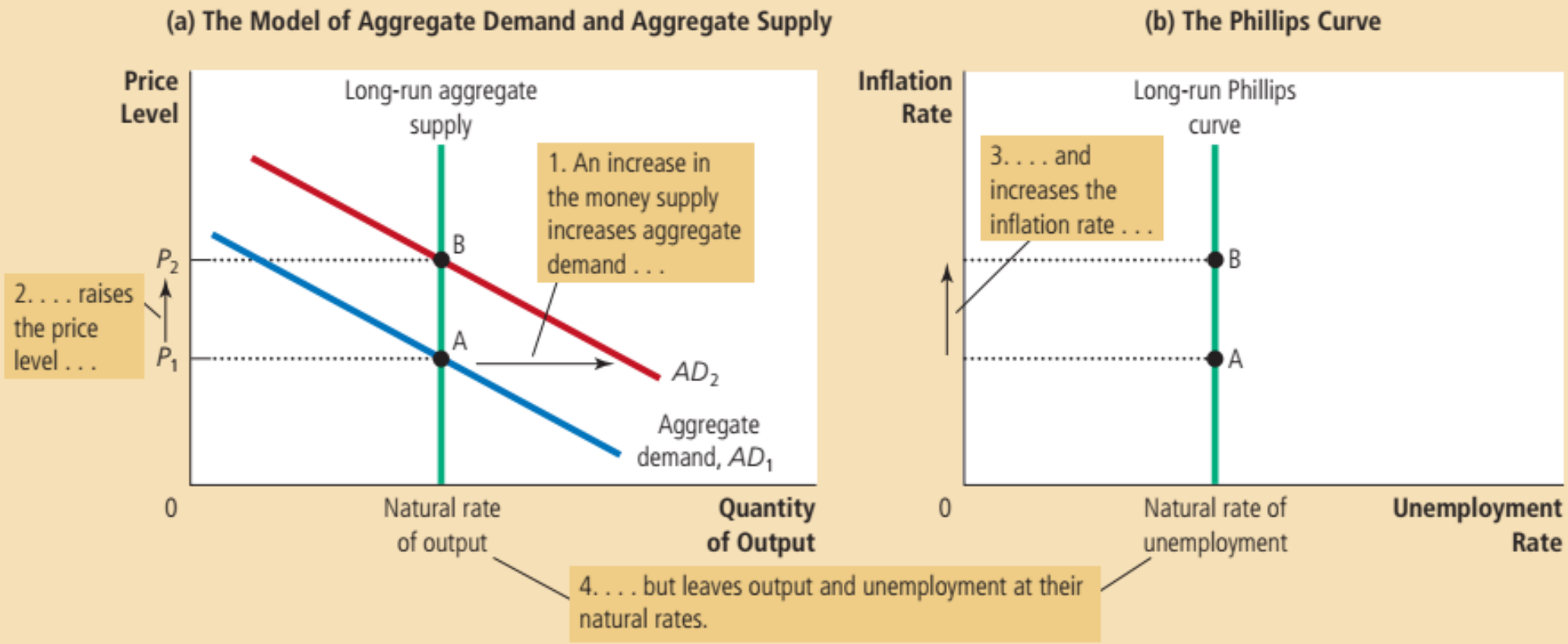
# Shifts in Phillips Curve

- Is the menu of choices going to be available always?
- Friedman and Phelps argued that acc. to classical theory, inflation is affected by Ms, and cannot affect real variables like output and unemployment.
- Unemployment will eventually trend towards its long run level.
- Long Run PC is vertical, analogous to LRAS.
- Another factor is supply shocks for shifts in PC.

Figure 4

## How the Long-Run Phillips Curve Is Related to the Model of Aggregate Demand and Aggregate Supply

Panel (a) shows the model of aggregate demand and aggregate supply with a vertical aggregate-supply curve. When expansionary monetary policy shifts the aggregate-demand curve to the right from $AD_1$ to $AD_2$, the equilibrium moves from point A to point B. The price level rises from $P_1$ to $P_2$, while output remains the same. Panel (b) shows the long-run Phillips curve, which is vertical at the natural rate of unemployment. In the long run, expansionary monetary policy moves the economy from lower inflation (point A) to higher inflation (point B) without changing the rate of unemployment.



# Long Run Phillips Curve

- LRPC depends on natural rate of unemployment, which depends on labor market structure.
- If policy reduces NAU, it shifts LRPC to the left, lower lower unemployment for every level of inflation.
- LRAS equivalently shifts to the right.
- Reconciling: role of expectations and sticky wages (similar to LRAS vs SRAS)

# Sticky Wages

- Sticky wages give rise to short run upward sloping AS curve; and short run downward sloping PC curve
- But sticky wages dont hold in long run, when households can revise their expectations of inflation
- Hence if inflation is kept high to keep unemployment low, eventually **expected inflation** rises, actual inflation equals expected inflation.

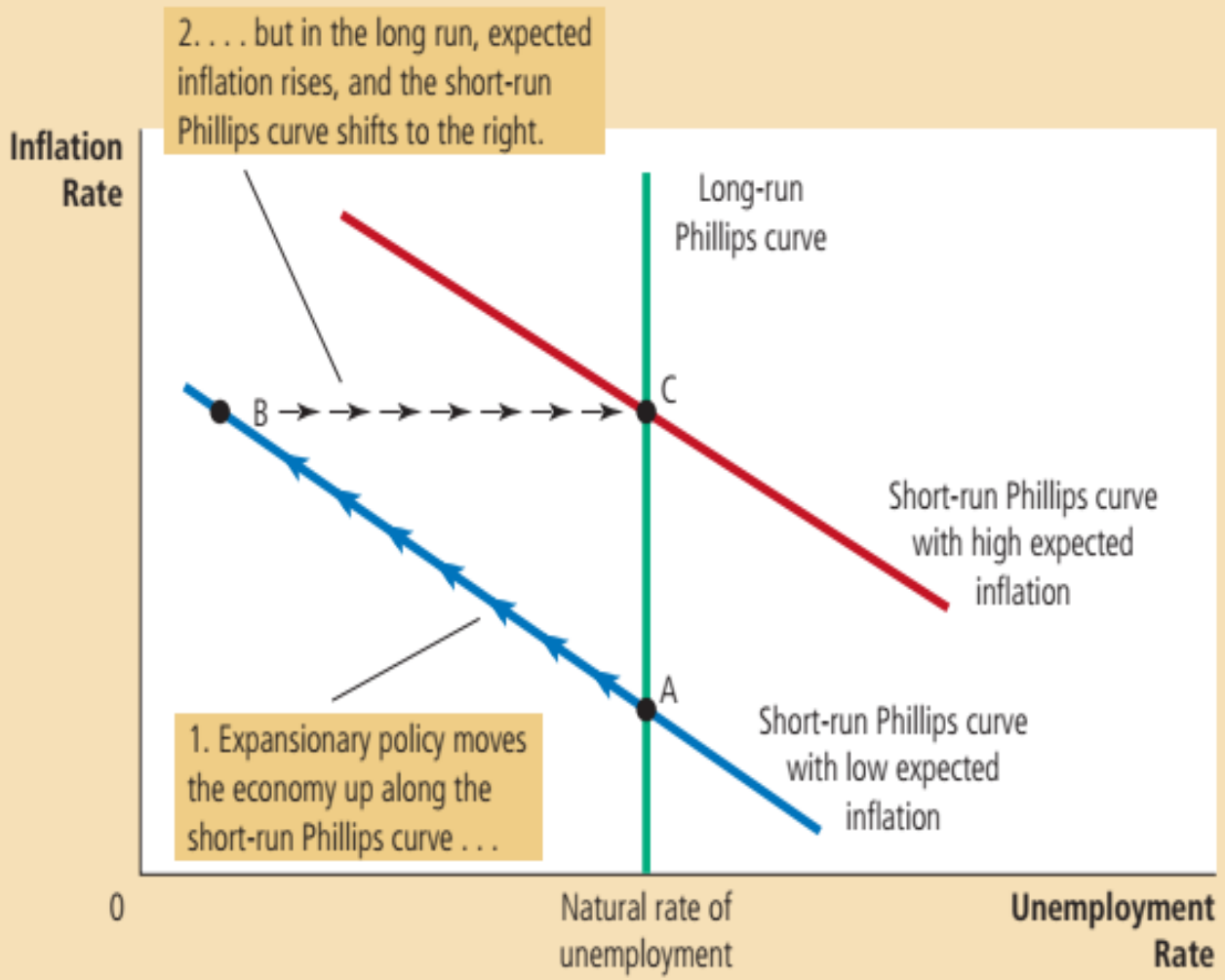
# The Short Run Phillips Curve

- Again, similar equation to AS for PC:
- Unemployment Rate = Natural Rate of Unemployment – a(Actual Inflation – Expected Inflation).

Figure 5

**How Expected Inflation Shifts the Short-Run Phillips Curve**

The higher the expected rate of inflation, the higher the short-run trade-off between inflation and unemployment. At point A, expected inflation and actual inflation are equal at a low rate, and unemployment is at its natural rate. If the Fed pursues an expansionary monetary policy, the economy moves from point A to point B in the short run. At point B, expected inflation is still low, but actual inflation is high. Unemployment is below its natural rate. In the long run, expected inflation rises, and the economy moves to point C. At point C, expected inflation and actual inflation are both high, and unemployment is back to its natural rate.



# Natural Experiment

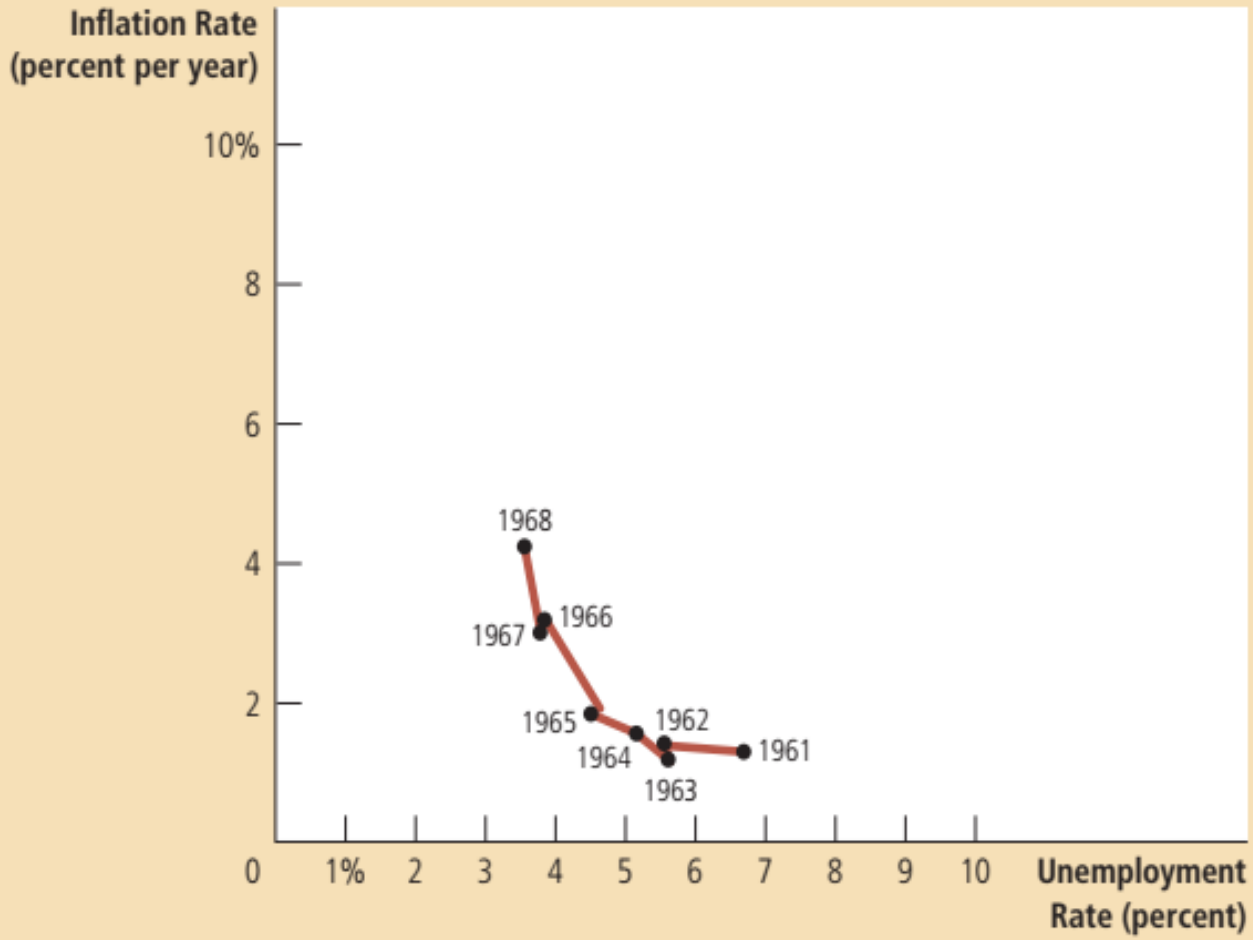



Figure 6

**The Phillips Curve in the 1960s**

This figure uses annual data from 1961 to 1968 on the unemployment rate and on the inflation rate (as measured by the GDP deflator) to show the negative relationship between inflation and unemployment.

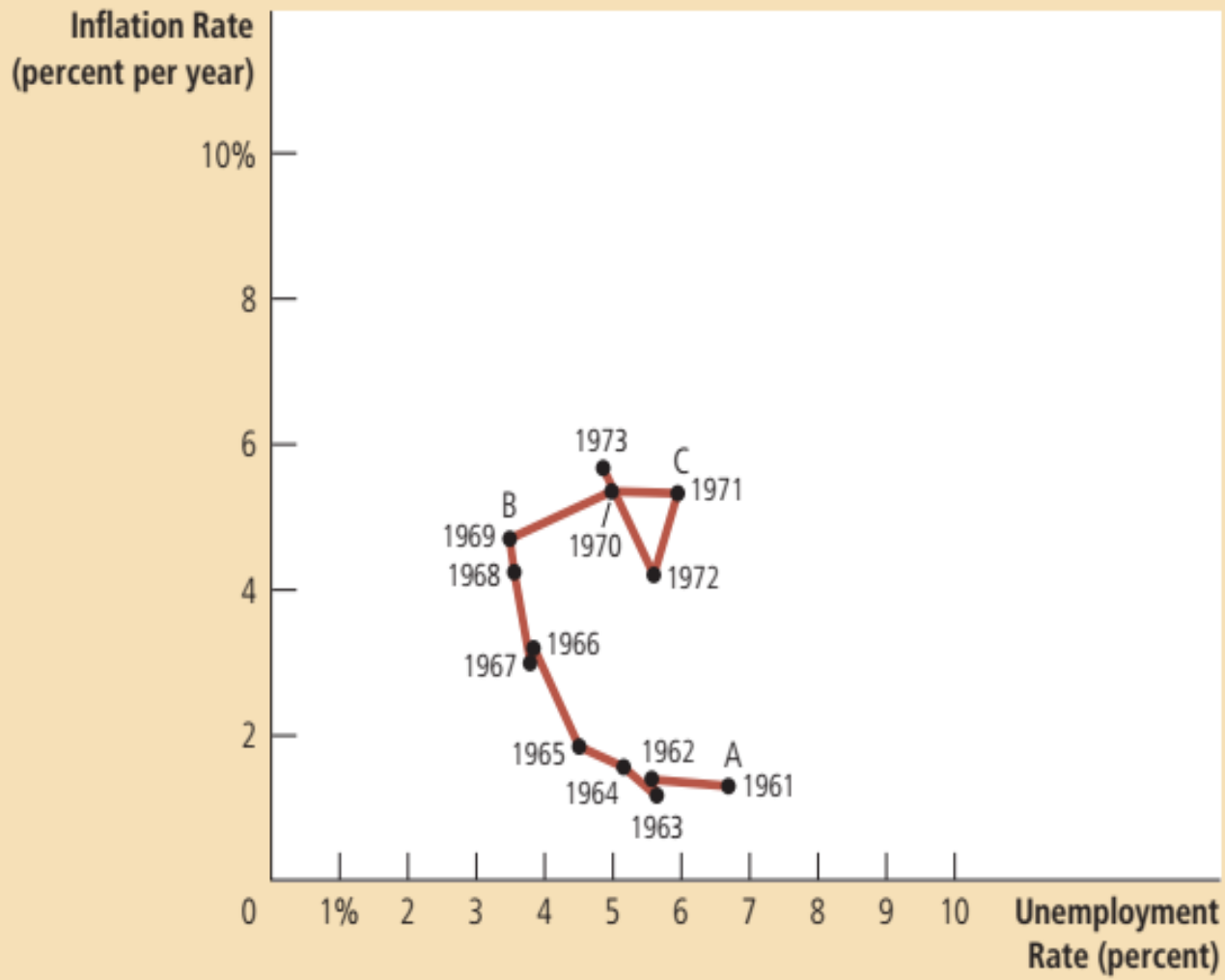
**Source:** U.S. Department of Labor; U.S. Department of Commerce.

Figure 7

**The Breakdown of the Phillips Curve**

This figure shows annual data from 1961 to 1973 on the unemployment rate and on the inflation rate (as measured by the GDP deflator). The Phillips curve of the 1960s breaks down in the early 1970s, just as Friedman and Phelps had predicted. Notice that the points labeled A, B, and C in this figure correspond roughly to the points in Figure 5.

**Source:** U.S. Department of Labor; U.S. Department of Commerce.



# Shifts in Phillips Curve: Role of Supply Shocks

Figure 8

**An Adverse Shock to Aggregate Supply**

Panel (a) shows the model of aggregate demand and aggregate supply. When the aggregate-supply curve shifts to the left from $AS_1$ to $AS_2$, the equilibrium moves from point A to point B. Output falls from $Y_1$ to $Y_2$, and the price level rises from $P_1$ to $P_2$. Panel (b) shows the short-run trade-off between inflation and unemployment. The adverse shift in aggregate supply moves the economy from a point with lower unemployment and lower inflation (point A) to a point with higher unemployment and higher inflation (point B). The short-run Phillips curve shifts to the right from $PC_1$ to $PC_2$. Policymakers now face a worse trade-off between inflation and unemployment.

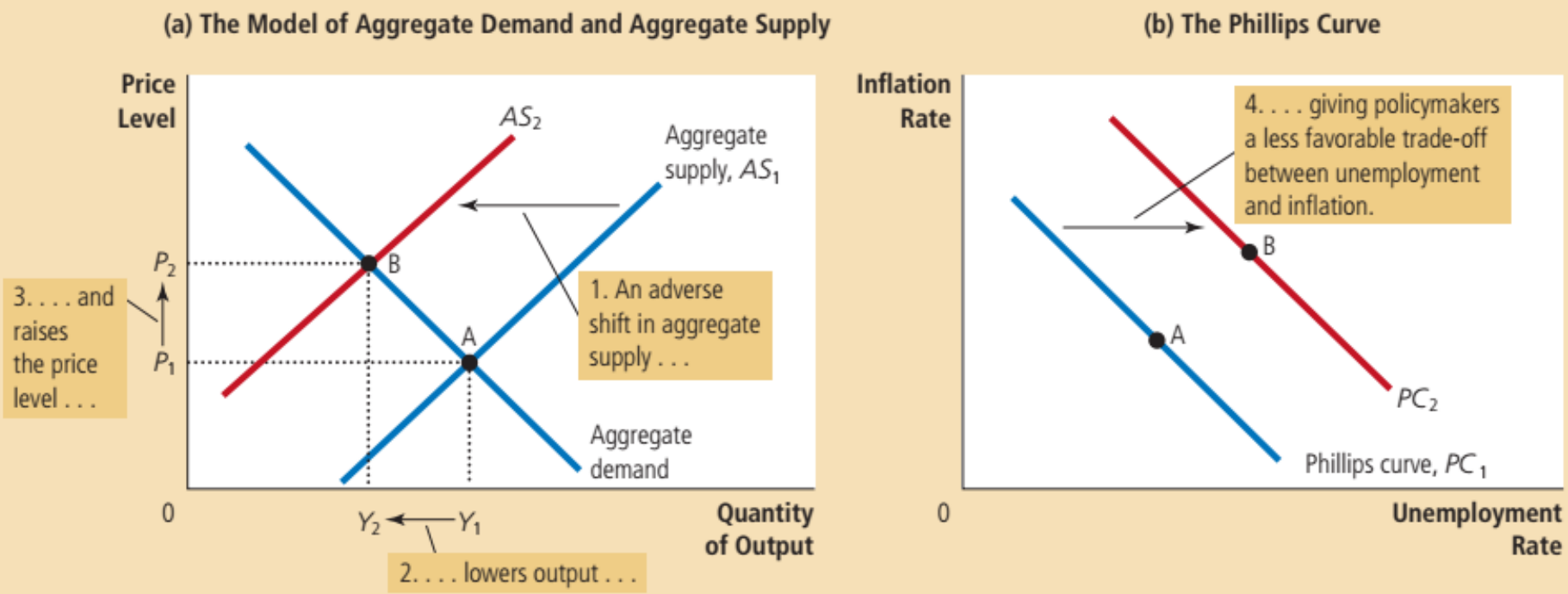
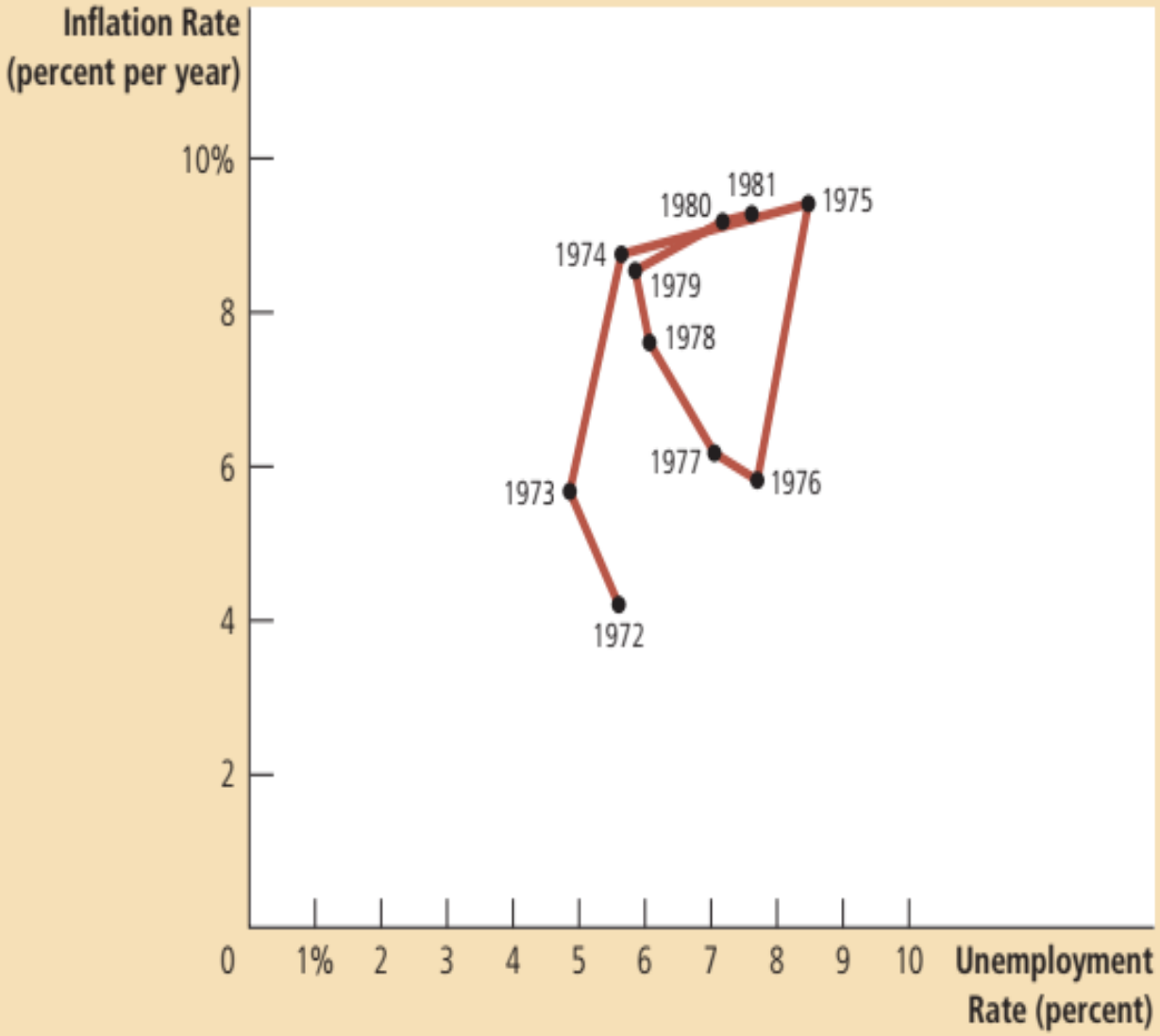


Figure 9

## The Supply Shocks of the 1970s

This figure shows annual data from 1972 to 1981 on the unemployment rate and on the inflation rate (as measured by the GDP deflator). In the periods 1973–1975 and 1978–1981, increases in world oil prices led to higher inflation and higher unemployment.

**Source:** U.S. Department of Labor; U.S. Department of Commerce.



# Cost of reducing Inflation

- Sacrifice Ratio: Number of percentage points of *annual* output lost in reducing inflation by one percentage point.
- So to reduce US inflation from 10 to 4%, they faced tradeoff of 30%!!
- Thankfully, expectations are not entirely *irrational*
- **Rational Expectations:** depends on credibility of message sent by Fed.

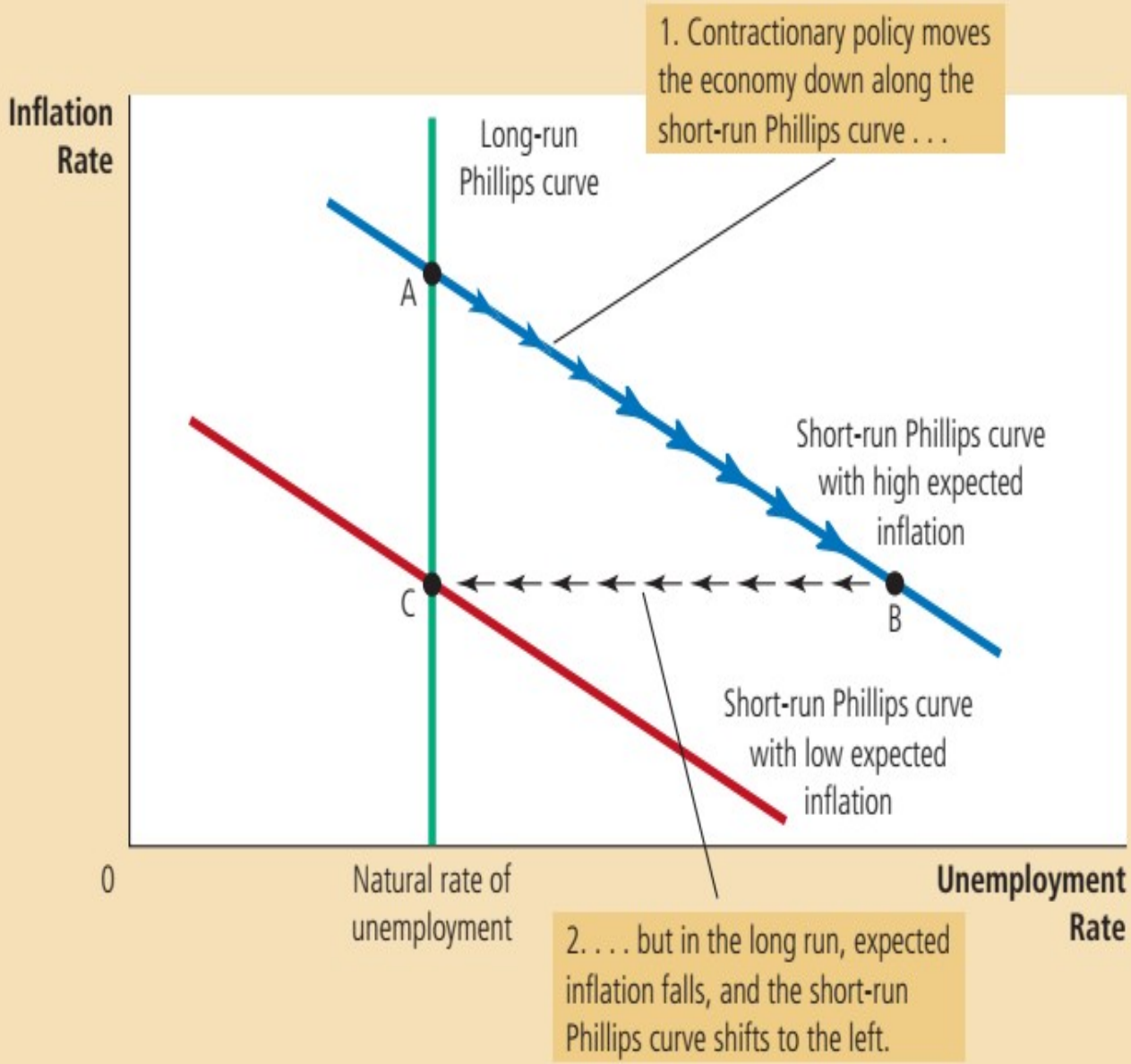



Figure 10

**Disinflationary Monetary Policy in the Short Run and Long Run**

When the Fed pursues contractionary monetary policy to reduce inflation, the economy moves along a short-run Phillips curve from point A to point B. Over time, expected inflation falls, and the short-run Phillips curve shifts downward. When the economy reaches point C, unemployment is back at its natural rate.

# RE and Volcker Dis-inflation

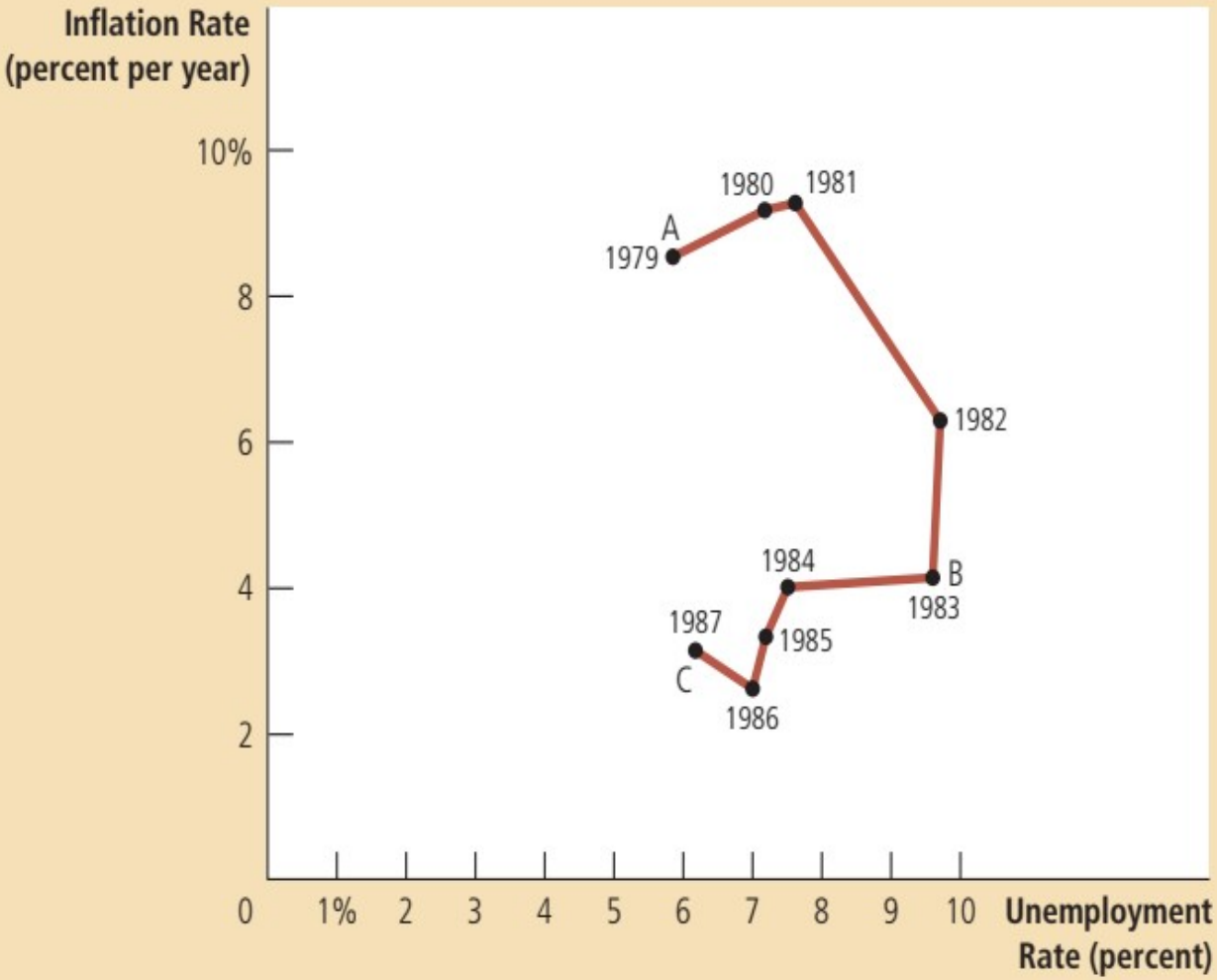
- The large costs associated with dis-inflation can be much lower
- If people believe inflation is going to be low next period, their expectations might also change accordingly.
- Hence, the ‘recession’ cost might be smaller

Figure 11

## The Volcker Disinflation

This figure shows annual data from 1979 to 1987 on the unemployment rate and on the inflation rate (as measured by the GDP deflator). The reduction in inflation during this period came at the cost of very high unemployment in 1982 and 1983. Note that the points labeled A, B, and C in this figure correspond roughly to the points in Figure 10.

**Source:** U.S. Department of Labor; U.S. Department of Commerce.



# Greenspan Era

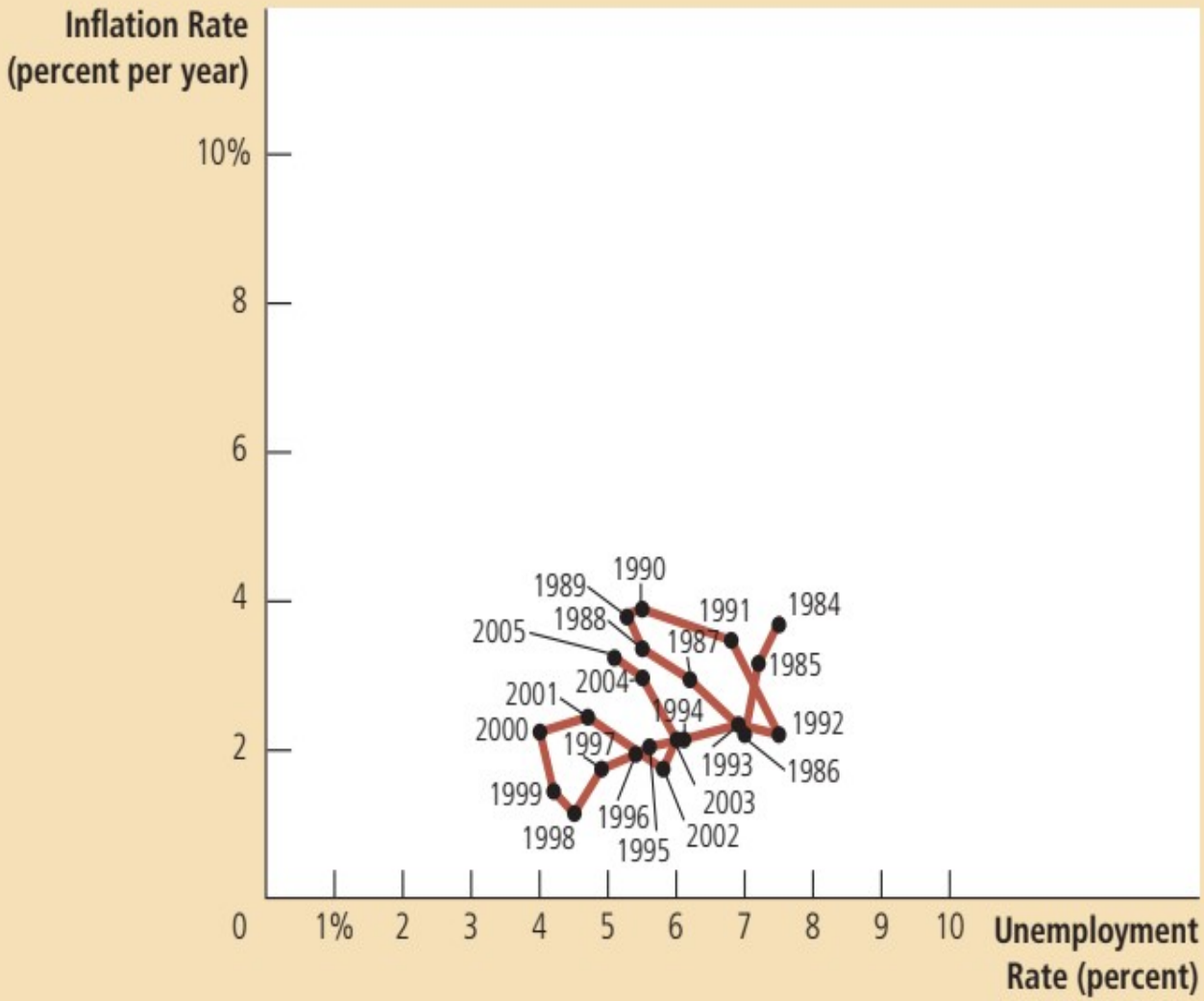



Figure 12

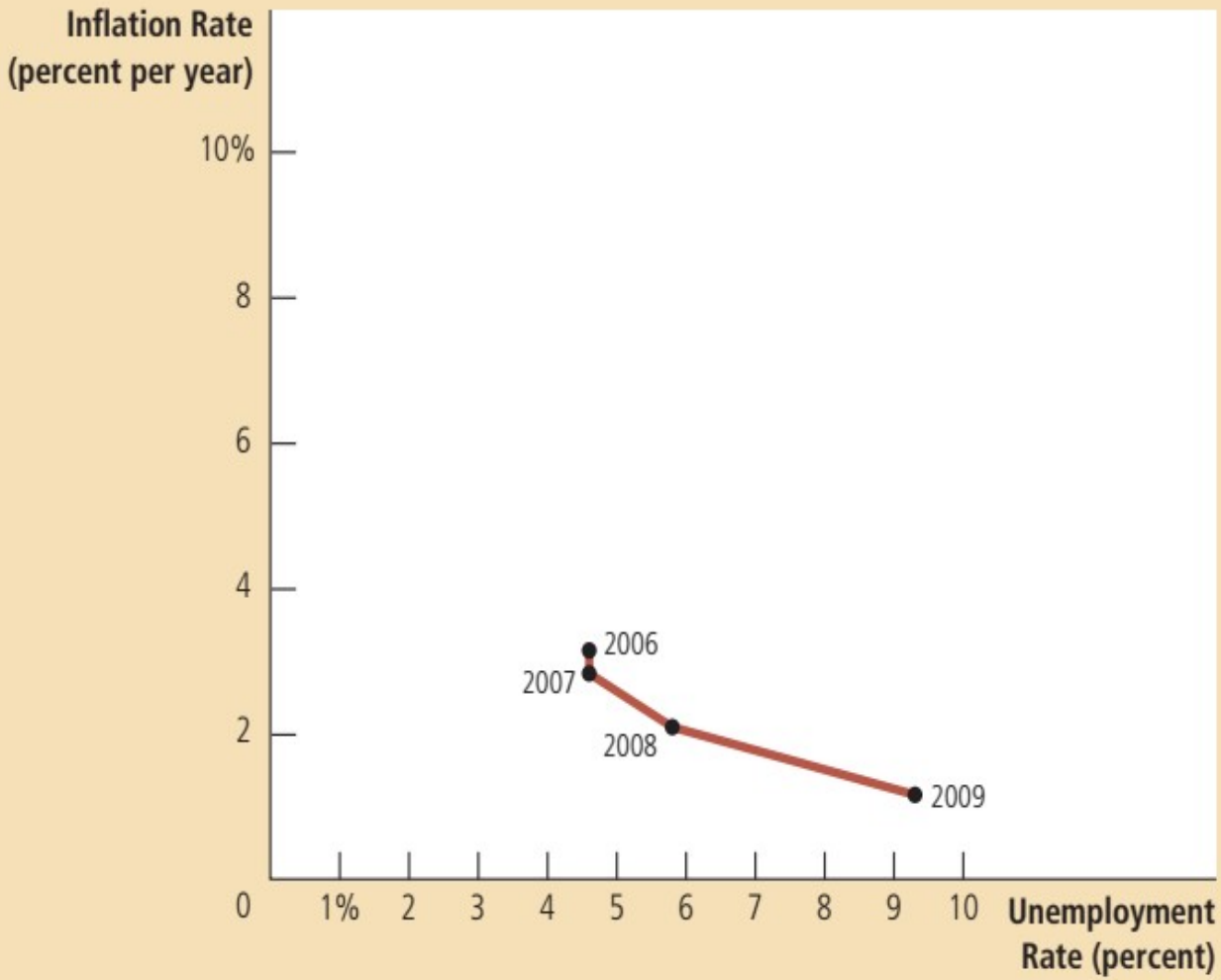
**The Greenspan Era**

This figure shows annual data from 1984 to 2005 on the unemployment rate and on the inflation rate (as measured by the GDP deflator). During most of this period, Alan Greenspan was chairman of the Federal Reserve. Fluctuations in inflation and unemployment were relatively small.

**Source:** U.S. Department of Labor; U.S. Department of Commerce.

Figure 13

**The Phillips Curve during the Recession of 2008–2009**

This figure shows annual data from 2006 to 2009 on the unemployment rate and on the inflation rate (as measured by the GDP deflator). A financial crisis caused aggregate demand to plummet, leading to much higher unemployment and pushing inflation down to a very low level.

**Source:** U.S. Department of Labor; U.S. Department of Commerce.